WORKED SOLUTION

STEP II 1997, Question 13

Probability and Statistics • Buffon's Needle

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Read the question before you go on. It is in the STEP Support Programme's question database:

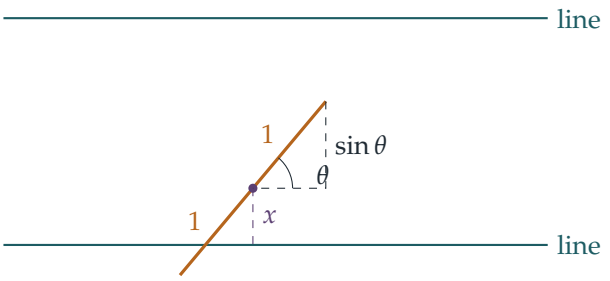
step.maths.org/questions/97-s2-q13

The key idea

This is Buffon's needle. Describe the needle's position by two independent random quantities: the distance of its centre from the nearest line, and the acute angle it makes with the lines. Once the centre is fixed, everything hinges on how far the needle reaches perpendicular to the lines, which is its half-length times the sine of that angle. Each part is then a question about a uniformly distributed angle.

Solution

We model "dropped at random" in the usual way: the distance X of the centre from the nearest line is uniform on $[0, 1]$ (the lines are 2 cm apart, so the nearest one is at most 1 cm away), and the acute angle θ between the needle and the lines is uniform on $[0, \frac{\pi}{2}]$, independently of X. (Taking the angle uniform on $[0, \pi)$ gives the same answers, since $\sin\theta$ has the same distribution either way.)



Each half of the needle has length 1, so the ends are at perpendicular distance $\sin\theta$ above and below the centre.

Part (i)

With the centre x from the nearest line, the needle crosses that line exactly when one end reaches past it, that is when

$$\sin\theta > x \iff \theta > \arcsin x,$$

since sin is increasing on $[0, \frac{\pi}{2}]$. As θ is uniform on $[0, \frac{\pi}{2}]$,

$$\mathrm{P}(\text{crosses nearest line} \mid X = x) = \frac{\frac{\pi}{2} - \arcsin x}{\frac{\pi}{2}} = 1 - \frac{2}{\pi}\arcsin x = \frac{2}{\pi}\arccos x.$$

Part (ii)

Suppose the needle crosses the nearest line, so $\theta > \arcsin x$. Measured along the needle, the crossing point is at distance $x/\sin\theta$ from the centre (the hypotenuse of a right-angled triangle with side x opposite θ). So the two pieces have lengths $1 + x/\sin\theta$ and $1 - x/\sin\theta$, and the shorter one is

$$S = 1 - \frac{x}{\sin\theta}.$$

As θ runs from $\arcsin x$ to $\frac{\pi}{2}$, S increases from 0 to $1 - x$, so S takes values in $[0, 1 - x]$. Conditioning on crossing, θ is uniform on $(\arcsin x, \frac{\pi}{2}]$, an interval of length $\frac{\pi}{2} - \arcsin x = \arccos x$. For $0 \le s \le 1 - x$,

$$S \le s \iff \frac{x}{\sin\theta} \ge 1 - s \iff \sin\theta \le \frac{x}{1-s} \iff \theta \le \arcsin\frac{x}{1-s},$$

where we used $1 - s > 0$ and $x/(1-s) \le 1$ (because $s \le 1 - x$), so the arcsine makes sense. Hence the cumulative distribution function of the shorter segment is

$$F(s) = \mathrm{P}(S \le s) = \begin{cases} 0, & s < 0, \\ \dfrac{\arcsin\left(\frac{x}{1-s}\right) - \arcsin x}{\arccos x}, & 0 \le s \le 1 - x, \\ 1, & s > 1 - x. \end{cases}$$

As a sanity check, the middle formula gives 0 at $s = 0$ and $(\frac{\pi}{2} - \arcsin x)/\arccos x = 1$ at $s = 1 - x$, so F is continuous.

Ask yourself Why is θ still uniform once we are told that the needle crosses the line?

Conditioning a uniform variable on lying in a sub-interval leaves it uniform on that sub-interval: for $\arcsin x \le a \le \frac{\pi}{2}$, $\mathrm{P}(\theta \le a \mid \theta > \arcsin x) = (a - \arcsin x)/(\frac{\pi}{2} - \arcsin x)$, which is linear in a. That is exactly what we used.

Part (iii)

The nearest line is at distance $X \le 1$, and every other line is at distance at least $2 - X \ge 1$. The needle reaches at most $\sin\theta \le 1$ from its centre, so it can cross a line other than the

nearest one only if $X = 1$ and $\theta = \frac{\pi}{2}$ exactly, which has probability zero. So the needle misses all the lines exactly when it misses the nearest line, and by part (i)

$$\mathrm{P}(\text{miss} \mid X = x) = \frac{2}{\pi} \arcsin x.$$

Since X is uniform on $[0, 1]$ (density 1), we average over x:

$$\mathrm{P}(\text{miss}) = \int_0^1 \frac{2}{\pi} \arcsin x \, \mathrm{d}x.$$

Integrating by parts with $u = \arcsin x$ and $\mathrm{d}v = \mathrm{d}x$,

$$\begin{aligned} \int_0^1 \arcsin x \, \mathrm{d}x &= \Big[x \arcsin x\Big]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} \, \mathrm{d}x \\ &= \frac{\pi}{2} - \Big[-\sqrt{1-x^2}\Big]_0^1 = \frac{\pi}{2} - 1. \end{aligned}$$

Therefore

$$\mathrm{P}(\text{needle misses all the lines}) = \frac{2}{\pi}\left(\frac{\pi}{2} - 1\right) = 1 - \frac{2}{\pi} \approx 0.363.$$

Checks

- Part (i) behaves sensibly at the ends: at $x = 0$ the centre is on the line and the probability is 1; as $x \to 1$ only a perpendicular needle reaches, and the probability tends to 0.
- The probability of crossing is $2/\pi \approx 0.637$, Buffon's classical answer $2\ell/(\pi d)$ with $\ell = d = 2$.
- A simulation of 10^6 random drops gave a miss rate of 0.3626, against $1 - 2/\pi = 0.3634$; simulated values of $F(s)$ for $x = 0.2, 0.5, 0.7$ agreed with the formula to about 0.002.

Common slips

- Forgetting to condition in part (ii): dividing by $\frac{\pi}{2}$ instead of $\arccos x$ gives a "distribution function" that never reaches 1.
- Taking the shorter segment as $1 - x\sin\theta$ or $1 - x/\cos\theta$: check which side of the triangle is x and which angle is θ.
- In part (iii), multiplying by $\frac{1}{2}$ or using X uniform on $[0, 2]$ without adjusting the crossing condition; the nearest-line distance is uniform on $[0, 1]$.

Beyond STEP

Buffon's needle (1777) is one of the first problems in geometric probability, and the appearance of π means that dropping needles gives a (slow) Monte Carlo estimate of π.

The cleanest modern treatment uses the linearity of expectation: the expected number of crossings is proportional to the needle's length, even for a bent needle ("Buffon's noodle"), which leads into integral geometry and the Crofton formula. Grimmett and Welsh, *Probability: An Introduction*, discusses joint distributions and geometric probability problems of this kind.

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